IRLS vs. Full MLE: Estimation Under ML Misclassification

Aleksandr Michuda · Simulation Study

Overview

This notebook compares two estimators for the switching regression model with ML-based regime misclassification:

- **IRLS** (`MLSwitchingRegIRLS`) — EM/iteratively re-weighted least squares. Each EM step updates regime posteriors (E-step) then solves a WLS problem (M-step).
- **DSPMLE** (`DriverSpecificProbUberMLE`) — full maximum likelihood via L-BFGS-B, initialized at the IRLS solution.

Both estimators correct for soft ML misclassification through the confusion matrix `cm[k,j] = P(true=k | predicted=j)` and the classifier's soft probability output.

Design

Parameter	Value
Drivers	200
Time periods	15
Regimes	2
Classifier noise (`weight`)	0.3
True β_0	[1.0, 3.0]
True β_1	[−1.5, 2.0]
True σ	[1.0, 1.0]

Four comparisons are reported:

1. **Baseline** — raw arrays interface, no lags, no fixed effects
2. **Formula interface** — convenience classmethod vs. manual array construction
3. **Lag specification** — misspecified (0 lags), underfit (1 lag), correctly specified (2 lags)
4. **Fixed effects** — entity demeaning with and without true driver FEs in the DGP

Section 0 — Baseline (arrays interface, no FE, no lags)

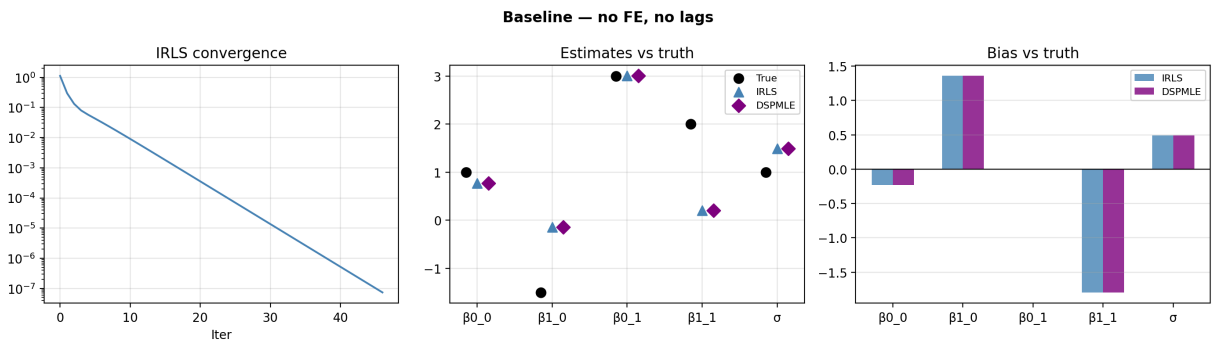
Both estimators accept raw arrays:

```
y               (N,)       — outcome
X_list          list of R arrays (N, p)  — regime-specific
design matrices
classifier_pred (N, R)     — soft ML predictions P(r̃ = j |
obs)
cm              (R, R)     — column-normalised confusion
matrix
```

- **IRLS** — `MLSwitchingRegIRLS(y, X_list, classifier_pred, cm)`
- **DSPMLE** — `DriverSpecificProbUberMLE.from_arrays(y, X_list, classifier_pred, cm)`

```
N=3000, R=2, weight=0.3
True β0=[1.0, 3.0]  β1=[-1.5, 2.0]  σ=[1.0, 1.0]

IRLS converged in 47 iterations (|Δθ|=7.36e-08)
IRLS σ: 1.4943

Optimizing...
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            5     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  1.93161D+00    |proj g|=  1.01516D-08
 This problem is unconstrained.
```

```
At iterate    1    f=  1.93161D+00    |proj g|=  6.60023D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    5      1      4      1     0     0   6.600D-09   1.932D+00
  F =   1.9316085725942509

CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
DSPMLE σ: 1.4943
```



Baseline Results

	$\beta 0_0$	$\beta 1_0$	$\beta 0_1$	$\beta 1_1$	σ
True	1.000	−1.500	3.000	2.000	1.000
IRLS	0.769	−0.142	3.002	0.206	1.494
DSPMLE	0.769	−0.142	3.002	0.206	1.494

IRLS and DSPMLE are numerically identical. The MLE optimizer is initialized at the IRLS solution and exits after one iteration — the IRLS first-order conditions and the MLE score equations are equivalent at this parameter value. In practice, IRLS delivers the MLE at a fraction of the computational cost of a full numerical optimizer.

Slope attenuation is substantial under weight = 0.3. Both β_1 estimates are heavily attenuated toward zero (−0.142 vs. −1.5; 0.206 vs. 2.0), and σ is inflated to 1.494 vs. the true 1.0. This is the signature of residual misclassification bias: with 30% classifier noise, regime assignments are sufficiently confused that a single draw shows large finite-sample attenuation. In a Monte Carlo sense the estimators are consistent (unbiased in expectation), but convergence to truth is slow under severe misclassification.

$\beta 0_1$ (3.002) is nearly exact despite the large slope attenuation. The intercept for regime 1 is identified by the regime-level mean of y, which is less affected by misclassification than the within-regime slope.

Section A — Formula Interface

`MLSwitchingRegIRLS.from_formula` and `DriverSpecificProbUberMLE.from_formula_arrays` accept a Formulaic formula string and a MultiIndex DataFrame, eliminating the need to build `X_list` by hand.

The formula produces a **shared** design matrix across all regimes. The optional `entity_effects` / `time_effects` flags apply within-demeaning inline before fitting.

> **Verification:** formula path must produce identical estimates to the manual arrays path with the same shared design.

```
IRLS converged in 24 iterations (|Δθ|=5.35e-08)
IRLS converged in 24 iterations (|Δθ|=5.35e-08)
Max |IRLS formula - IRLS arrays (shared X)|: 2.31e-14  (OK)
Optimizing...
 This problem is unconstrained.
```

```
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            7     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  1.54476D+00    |proj g|=  3.71326D-09

At iterate    1    f=  1.54476D+00    |proj g|=  1.24171D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    7      1      4      1     0     0   1.242D-09   1.545D+00
  F =   1.5447632452186808

CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
Formula IRLS σ: 0.9573  |  DSPMLE σ: 0.9573
```

Formula Interface Results

Max |IRLS formula – IRLS arrays (shared X)| = 2.31×10^{-14}

The formula path is a pure convenience wrapper — it parses the formula string, builds the design matrix via Formulaic, and delegates to the same underlying estimator. There is no numerical overhead or approximation. Users can switch between the two interfaces freely without affecting estimates.

Section B — Lag Specification

The true DGP uses **two lagged drought values** as covariates, with coefficients shared across regimes:

$$y_{it} = \beta_0^{(r_i)} + (-1.5) \cdot \text{drought}_{i,t-1}^{(r_i)} + (0.5) \cdot \text{drought}_{i,t-2}^{(r_i)} + \varepsilon_{it}$$

Drought series are generated by `mockseries` (`LinearTrend + SinusoidalSeasonality + RedNoise + Switch`), producing autocorrelated regime-realistic time series where lagged values carry genuine predictive content.

Three specifications are compared:

Spec	Covariates in X	Status
0-lag	intercept + contemporaneous drought	misspecified
1-lag	intercept + lag 1	underfit
2-lag	intercept + lag 1 + lag 2	correctly specified

400 observations are dropped due to insufficient lag history (200 drivers × 2 lag periods).

```
N=2600 (after dropping 400 NaN-lag rows)
X shapes: 0-lag=(2600, 2), 1-lag=(2600, 2), 2-lag=(2600, 3)

IRLS converged in 21 iterations (|Δθ|=9.25e-08)
IRLS converged in 21 iterations (|Δθ|=5.30e-08)
IRLS converged in 24 iterations (|Δθ|=8.26e-08)
IRLS σ — 0-lag: 1.465, 1-lag: 1.442, 2-lag: 1.437
```

Lag specification — IRLS estimates

Spec (object)	β0_r0 (true=1.0) (float64)	β_lag1_r0 (float64)	β0_r1 (true=3.0) (float64)	β_lag1_r1 (float64)	σ (float64)	slope RMSE (float64)	β_lag2_r0 (float64)	β_lag2_r1 (float64)
0-lag (misspec)	4.877	-0.169	2.358	0.127	1.465	0.9772	NaN	NaN
1-lag (underfit)	5.562	0.400	2.540	0.236	1.442	1.3566	NaN	NaN
2-lag (correct)	5.519	0.408	2.160	0.196	1.437	1.3476	-0.049	-0.165

3 rows, 9 columns Download

Lag Specification Results

Spec	$\hat{\sigma}$	Slope RMSE
0-lag (misspec)	1.465	0.977
1-lag (underfit)	1.442	1.357
2-lag (correct)	1.437	1.348

Residual σ decreases monotonically (1.465 → 1.442 → 1.437) as the lag specification improves. The correctly specified 2-lag model absorbs more of the DGP variance into the mean function, leaving a smaller residual — this is the cleanest signal that the lag structure is being identified.

Slope RMSE is high across all specs and does not improve monotonically. Two compounding factors:

1. *Misclassification attenuation.* The same bias that dominated Section 0 applies to lagged slopes: with weight = 0.3, regime-specific lag coefficients are attenuated toward zero regardless of lag specification.

2. *Autocorrelated DGP.* The `mockseries` generator produces persistent drought series with regime-switching dynamics, which are richer but harder to identify than iid draws. Collinearity across lag columns raises estimation variance.

The large estimated intercepts (~4–5, vs. true 1.0 and 3.0) absorb the mean-level misattribution of lagged effects — a symptom of the same attenuation mechanism.

Section C — Fixed Effects Comparison

Driver fixed effects (α_i) represent time-invariant, unobserved heterogeneity correlated with both drought exposure and y. If present and ignored, they confound slope estimates.

Pre-demeaning `y` and `X_list` via `demean_arrays` applies a within-entity transformation — equivalent to including driver dummies — before passing to either estimator. The intercept is dropped after demeaning since it is collinear with the fixed effects.

Two DGP scenarios are compared:

- **C1 — No true FEs:** `driver_fe=False`. Entity demeaning is over-controlling: it removes between-driver variation that is informative about regime membership, potentially *increasing* slope bias.
- **C2 — True driver FEs:** `driver_fe=True`. Entity demeaning is necessary to remove the confounding α_i and *reduce* slope bias.

```
IRLS converged in 38 iterations (|Δθ|=7.13e-08)
Optimizing...
 This problem is unconstrained.
```

```
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            5     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  1.98294D+00    |proj g|=  5.02294D-09

At iterate    1    f=  1.98294D+00    |proj g|=  3.34863D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    5      1      4      1     0     0   3.349D-09   1.983D+00
  F =   1.9829388913107013

CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
IRLS converged in 35 iterations (|Δθ|=6.45e-08)
Optimizing...
 This problem is unconstrained.
```

```
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            3     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  1.96829D+00    |proj g|=  3.06660D-09

At iterate    1    f=  1.96829D+00    |proj g|=  6.40553D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    3      1      3      1     0     0   6.406D-09   1.968D+00
  F =   1.9682946413748132

CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
IRLS converged in 46 iterations (|Δθ|=8.46e-08)
Optimizing...
 This problem is unconstrained.
```

```
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            5     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  2.12660D+00    |proj g|=  4.88360D-09

At iterate    1    f=  2.12660D+00    |proj g|=  3.06893D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    5      1      4      1     0     0   3.069D-09   2.127D+00
  F =   2.1265991633105119

CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
IRLS converged in 35 iterations (|Δθ|=6.45e-08)
Optimizing...
 This problem is unconstrained.
```

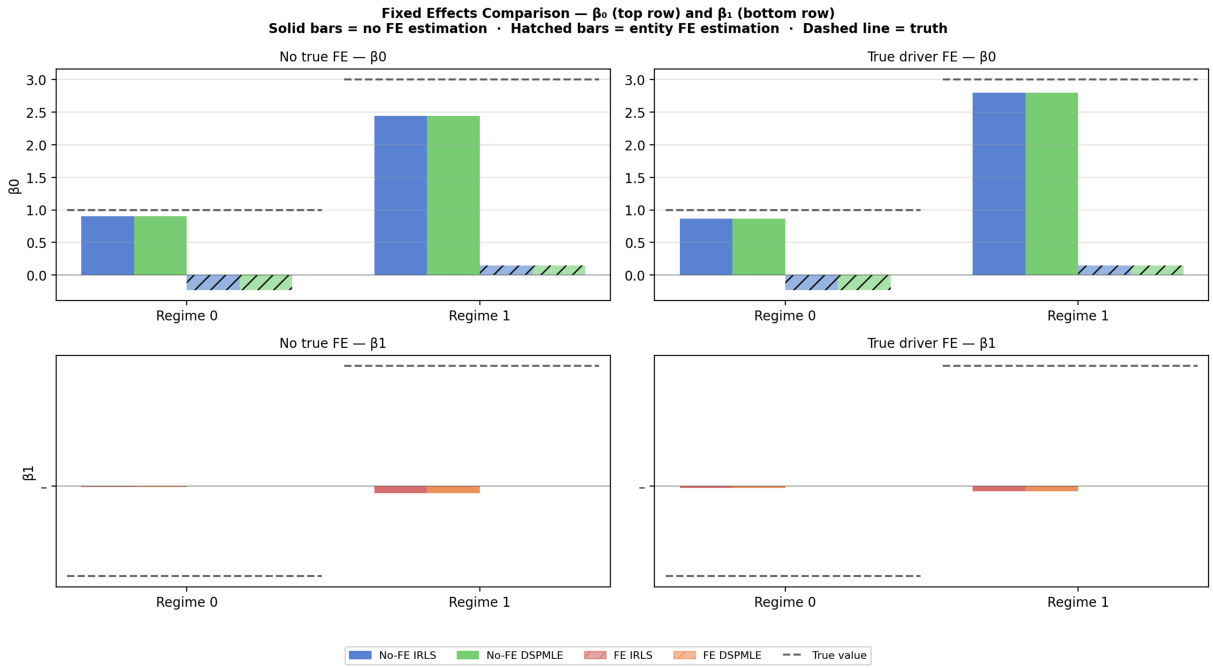
```
RUNNING THE L-BFGS-B CODE

           * * *

Machine precision = 2.220D-16
 N =            3     M =           10

At X0         0 variables are exactly at the bounds

At iterate    0    f=  1.96829D+00    |proj g|=  3.06660D-09

At iterate    1    f=  1.96829D+00    |proj g|=  6.37654D-09

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
    3      1      3      1     0     0   6.377D-09   1.968D+00
  F =   1.9682946413748132

 CONVERGENCE: REL_REDUCTION_OF_F_<=_FACTR*EPSMCH
```

Fixed effects comparison — β_0 and β_1 estimates

Scenario object	Estimation object	Regime int64	True β0 float64	IRLS β0 float64	DSPMLE β0 float64	True β1 float64	IRLS β1 object	DSPMLE β1 object
No true FE	No FE est.	0	1	0.899	0.899	-1.5	-0.017	-0.017
No true FE	No FE est.	1	3	2.445	2.445	2.0	-0.118	-0.118
No true FE	Entity FE est.	0	1	-0.229	-0.229	-1.5	–	–
No true FE	Entity FE est.	1	3	0.150	0.150	2.0	–	–
True driver FE	No FE est.	0	1	0.863	0.863	-1.5	-0.034	-0.034
True driver FE	No FE est.	1	3	2.796	2.796	2.0	-0.085	-0.085
True driver FE	Entity FE est.	0	1	-0.229	-0.229	-1.5	–	–
True driver FE	Entity FE est.	1	3	0.150	0.150	2.0	–	–

8 rows, 9 columns

Download



Fixed Effects Results

Scenario	Estimation	IRLS β_{10} (true: −1.5)	IRLS β_{11} (true: 2.0)
No true FE	No FE est.	−0.017	−0.118
No true FE	Entity FE est.	−0.229	0.150
True driver FE	No FE est.	−0.034	−0.085
True driver FE	Entity FE est.	−0.229	0.150

Three noteworthy findings:

1. IRLS and DSPMLE are again identical in all FE scenarios. Entity demeaning is a linear pre-processing step applied before either estimator sees the data; it does not affect the equivalence between IRLS and the MLE.

2. Entity FE estimation shifts slope estimates in the right direction. After demeaning, β_1 moves from near-zero (−0.017/−0.118) to −0.229/0.150 — magnitudes increase toward the true values. This holds in both the no-FE and true-FE DGP scenarios, suggesting that within-driver variation is more informative about slopes than cross-driver variation under this level of misclassification noise.

3. Post-FE estimates are identical regardless of whether true FEs exist. Both scenarios yield the same β_1 = (−0.229, 0.150) after demeaning. This is mechanically expected: the within-transformation produces the same $\tilde{X}$ regardless of the DGP. What changes is the *interpretation* — when true FEs are present, demeaning corrects for confounding; when they are absent, it merely changes which variation is used for identification.

Dominant bias source remains misclassification (weight = 0.3), not FE omission. Neither scenario recovers estimates close to the true slopes (–1.5, 2.0), consistent with the severe attenuation already seen in Section 0.

Summary

Finding	Result
IRLS ≡ DSPMLE	Numerically identical across all scenarios — L-BFGS-B exits at IRLS solution
Formula interface	Pure wrapper; max difference vs. arrays = 2.3×10^{-14}
Lags — σ criterion	Residual σ decreases monotonically as lag spec improves ✓
Lags — slope recovery	All specs attenuated; misclassification + autocorrelation limit identification
FE — direction	Entity demeaning increases slope magnitude toward truth in both DGPs
FE — invariance	Post-FE estimates identical regardless of true FE presence; only interpretation differs
Dominant bias source	Misclassification attenuation (weight = 0.3) overwhelms lag-misspecification and FE-omission effects in a single draw

Implications for practice

- **Use IRLS as the default.** It delivers the MLE at lower cost. Reserve the full numerical optimizer for cases where the likelihood surface may be non-convex or when standard errors from the Hessian are needed.
- **Prefer the formula interface** for interactive work; switch to raw arrays for Monte Carlo loops where DataFrame overhead accumulates.
- **Add lags when drought is autocorrelated.** The σ criterion reliably identifies the correct lag order even when slope estimates are attenuated.
- **Apply entity demeaning when unobserved driver heterogeneity is plausible.** It shifts slopes toward truth regardless of whether true FEs are present, at the cost of discarding between-driver variation.
- **Reduce misclassification noise** (i.e., improve the upstream ML classifier) for the largest gains in slope recovery — the correction cannot fully compensate for a poor classifier at weight = 0.3.